Where does the infill housing go? Addition versus division

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March 24, 2026

Overview

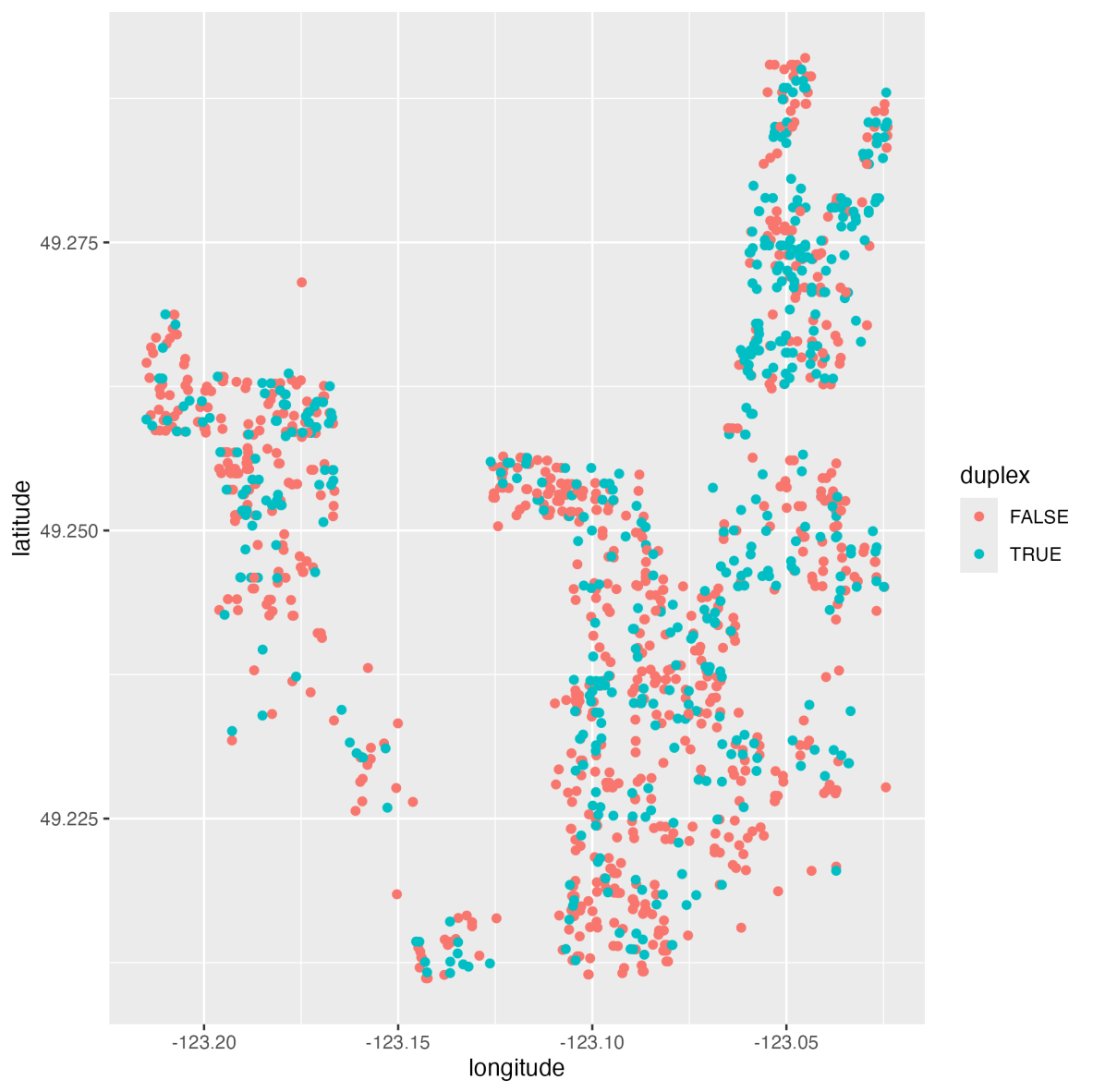
- Introduce a puzzle regarding upzoning for "infill housing"
  - New density allowances in "single family" zones *should* be most valuable in high-priced areas
  - But takeup of infill options leans towards lower-priced areas
- Resolve the puzzle with a model
  - Distinguish "addition" (more square feet) from "division" (more units)
- Present preliminary evidence on the model's performance
- Data: upzonings in Vancouver, Portland, and Minneapolis
- Empirical contributions:
  - Model is right: division more attractive in worse locations
  - Mixed evidence that addition-heavy upzonings go to relatively better locations

Zoning deadweight loss greatest in best locations

- Glaeser and Gyourko (2003) and Brueckner and Singh (2008): compare marginal price to cost /sqft
- Cities' natural shape: tallest, densest use closest to downtown
- For structure MC = price, need a very tall building downtown
- For land and structure bang-for-buck equality, need low density in suburbs
- Policy guidance re: upzoning to improve affordability:

The model in this report identifies the lots most ripe for redevelopment under the specified density scenarios, where the gains for converting from the exist- ing single family properties are highest, as the size of the wedge. Consistent with the basic model in urban economics, these are areas where land values for single family lots are highest. In Metro Vancouver, the City of Vancouver (especially its west side), the North Shore, Richmond and Burnaby stand out. These results correspond well with expectations that high amenity ar- eas closest in to the metropolitan centre would receive the greatest attention for higher density redevelopment if it were enabled.

Thus when Vancouver allowed 2 homes vs 1 per lot. . . Oh no!



Not just Vancouver duplex where infill goes to less expensive land

- Vancouver laneway (Davidoff, Pavlov, Somerville, 2022)
- ADUs in California (Marantz-Elmendorf and Kim, 2023; Brueckner and Thomaz, 2024)
- Duplexes and triplexes in Seattle (Huang et al, 2023)
- Portland's Residential Infill Project (RIP) (Dong and Hansz, 2019)
- Maybe we need a better model of infill location

A model of infill location

- "primitives"
  - $p(s, \theta)$: price per square foot of structure
    - Size $s = \frac{f}{n}$
    - f: floor area
    - n: number of units
    - Location θ
    - Lot size fixed at 1 unit of land
    - Assume: $p_s > 0$, $p_\theta > 0$.
    - Reduced form sorting (single family!): $p_{s\theta} > 0$
  - Profit from raw land development $\pi = f\left[p(s, \theta) - c\right]$
    - c: cost per sqft of structure, constant for now
    - f not necessarily freely chosen

Upzoning comparative statics addition versus division

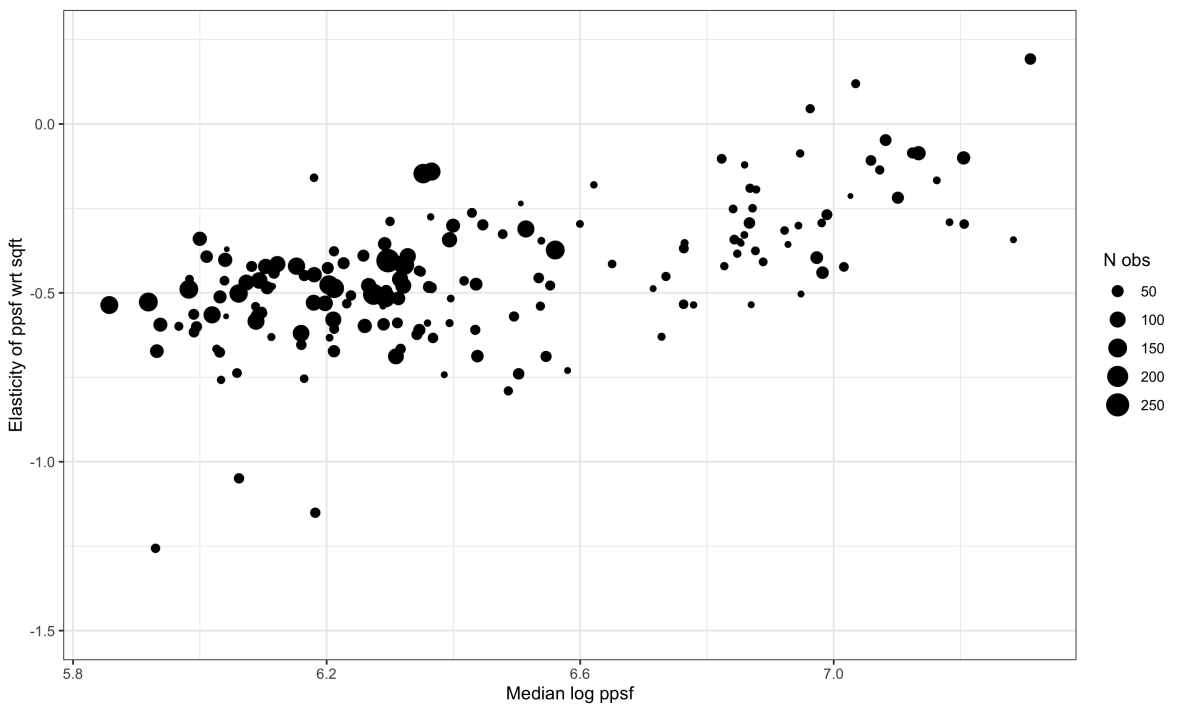
- Addition: allow extra f
  - $\frac{\partial \pi}{\partial f} = p - c + sp_s$
  - $\frac{\partial^2 \pi}{\partial f \partial \theta} = p_\theta + sp_{s\theta} > 0$: more profitable in better locations
- Division: allow more n
  - $\frac{\partial \pi}{\partial n} = -s^2 p_s$
  - $\frac{\partial^2 \pi}{\partial n \partial \theta} = -s^2 p_{s\theta} < 0$: more profitable in worse locations

Upzonings in Vancouver

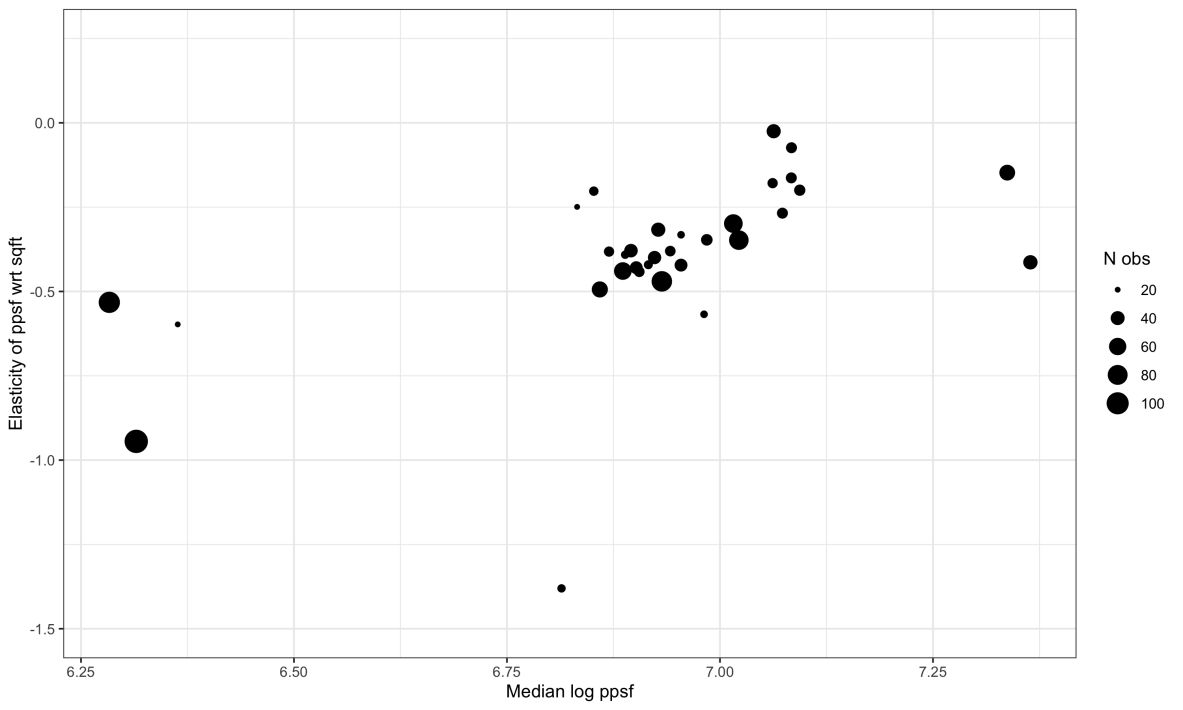
- Predominant Zoning: R1-1 now, used to be several R zones
- Baseline was single family detached homes
- 1980s-2004: Basement suites permitted citywide
- 2009: laneway homes permitted in single family homes
  - Add square feet, but must be rented, lose garage space
  - Laneway allowed .16-.25 FSR
- Sept 2018: duplexes allowable throughout R zones
  - No extra square feet (FSR .6-.7), but can sell 2 units
  - Generally *can't* add a laneway
  - Hence division and *negative* addition
- 2023: multiplexes allowed
  - 4-6 units permitted depending on lot size
    - Much more attractive on wider lots
    - Roughly 50% of Vancouver lots 33' standard
  - FSR goes from .7 to 1.0
  - Hence more division, more addition than duplex

Is it true that $p_{s\theta} > 0$?

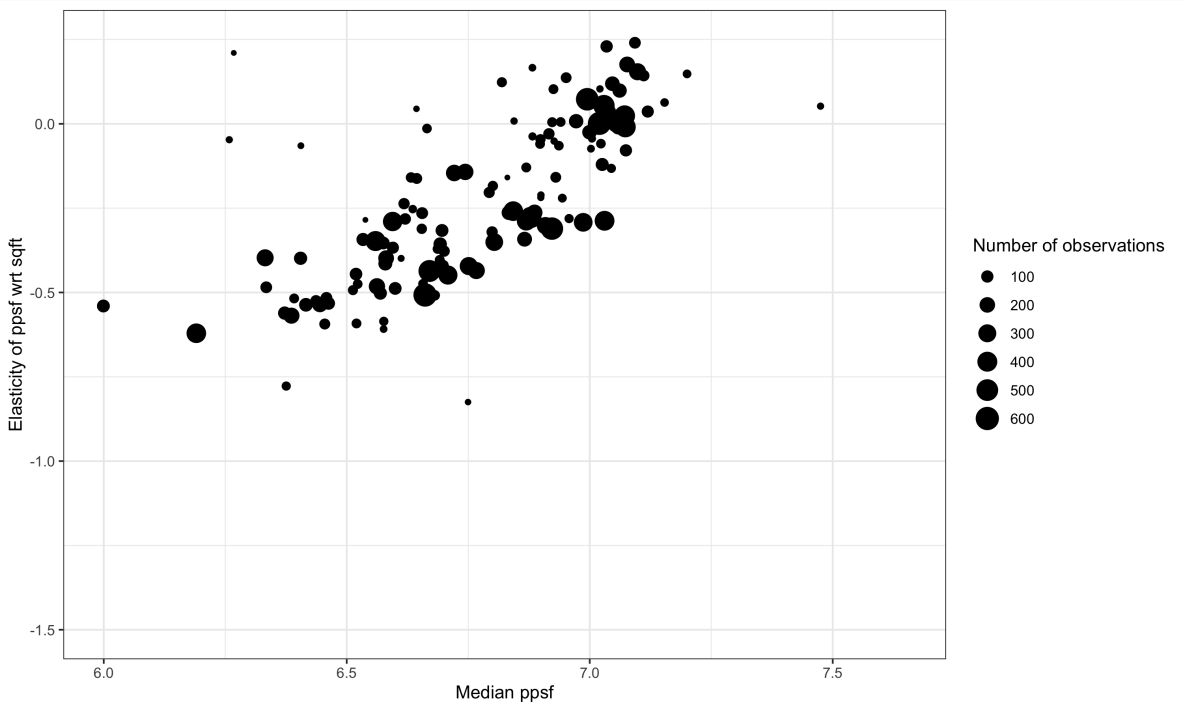
- BCA Single Family sales 2023-2024, Metro Vancouver
- Observation: a neighbourhood
- Elasticity < -1 means $p_s < 0$, > 0 means want lot assembly
  - Both implausible



Price per square foot and elasticity: duplexs



Price per square foot and elasticity: condos
Constant elasticity not a good fit



Spec vs custom: single vs not

- Spec(ulative): developer buys, builds, sells to market
- Custom: rich guy buys, hires builder, holds property
- Hard to do custom other than single family
- Exercise: match CoV new build permit to subsequent sale, 2017-2021:
  - Duplex 41 no, 77 yes
  - Single 1884 no,706 yes
- Custom likely fancier and not observed in sales data
- Elasticity hence likely biased down, more so West Side
- Note: can't do recent time slope of custom share: mechanical

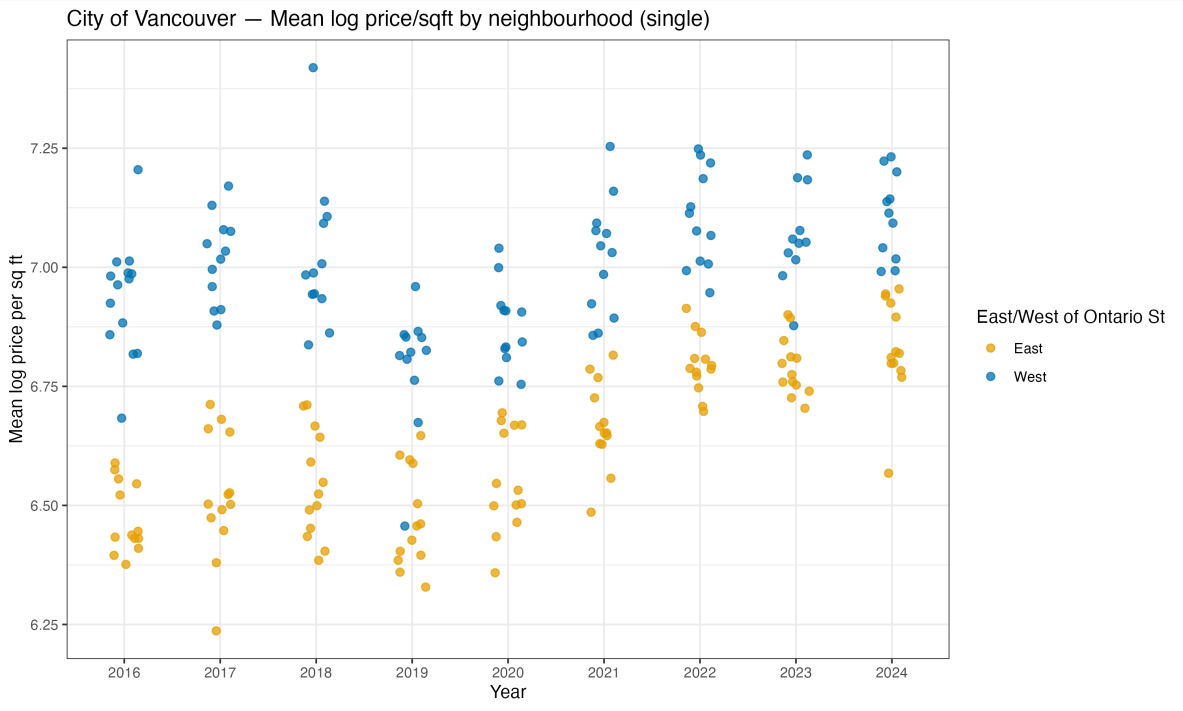
Not surprisingly, $p_{s\theta}^{condo} < p_{s/theta}^{duplex} < p_{s\theta}^{sf}$

- Condos built to maximize $p - c$
- Singles too big
- Duplexes presumably ease waste of space
- Elasticity < -1 means $p_s < 0$, > 0 means want lot assembly
    - Both implausible

City of Vancouver unit regressions by type

Dependent Variable:	logppsf		
Model:	(1)	(2)	(3)
Variables			
logsqft	-0.571***	-0.651***	-0.079***
	(0.002)	(0.012)	(0.002)
Fixed-effects			
year	Yes	Yes	Yes
NEIGHBOURHOOD	Yes	Yes	Yes
Fit statistics			
Observations	99,761	9,826	201,358
R^2	0.77528	0.58459	0.69394
Adjusted R^2	0.77480	0.57858	0.69374

Note dramatic price compression over time (foreign/spec?)
Dot is a neighbourhood x year



Duplex analysis for Vancouver, 2019-2023 (post-duplex, pre-multiplex)

- Focus on 33' lots (majority of Vancouver)
- Choice: duplex - single family alone premium
  - Laneway complicated (discussion)
- Comes from neighbourhood single family price per sqft and elasticity

Correlations: minimum 20 single family sales observations

	elasticity	intercept	duplex_premium	shareDuplex
elasticity	1.00	-0.99	-0.89	-0.11
intercept	-0.99	1.00	0.94	0.08
duplex_premium	-0.89	0.94	1.00	0.01
shareDuplex	-0.11	0.08	0.01	1.00

	elasticity	intercept	duplex_premium	shareDuplex
elasticity	1.00	-0.99	-0.84	-0.15
intercept	-0.99	1.00	0.91	0.16
duplex_premium	-0.84	0.91	1.00	0.14
shareDuplex	-0.15	0.16	0.14	1.00

	elasticity	intercept	duplex_premium	shareDuplex
elasticity	1.00	-0.99	-0.91	-0.22
intercept	-0.99	1.00	0.95	0.21
duplex_premium	-0.91	0.95	1.00	0.17
shareDuplex	-0.22	0.21	0.17	1.00